Practice problems #27

1. Find bases for Row$A$ and Row$B$, where A is row equivalent to B:

$$A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 0 & 5 & -7 & 8 \\ 0 & 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

2. Suppose that a nonhomegeneous system of six linear equations in eight unknowns has a solution, with two free variables. Is it possible to change some constants on the equations' right-hand-sides to make the new system inconsistent? Explain.

3. Let A be an $m \times n$ matrix. Which of the subspaces Row A, Col A, Nul A, Row A^T, Col A^T, and Nul A^T are in $\mathbf{R^m}$ and which are in $\mathbf{R^n}$? How many distinct subspaces are in this list?

4. Let A be an $m \times n$ matrix. Prove that the equation $Ax = b$ has a solution for all $b \in \mathbf{R^m}$ if and only if the equation $A^T x = 0$ has only the trivial solution.

Answers or Hints are on page 2.

#1: Easy. One answer works for both Row$A$ and Row$B$.

#2: No. (nullity = 2, so rank = 6, hence??)

#3: Subspaces of $\mathbf{R^n}$ are Row A and Nul A; Subspaces of $\mathbf{R^m}$ are Col A and Nul A^T; four distinct subspaces.

#4: Hint: rank A^T = dim Col A^T (by definition) = dim Row$A$ (showed in class)